

Self-educated and of humble origins, Benjamin Franklin was an inventor, scientist, printer, and politician. Through his scientific experiments, he made significant advances in our understanding of electricity and produced several key inventions. He is also revered as one of the Founding Fathers of the United States.

The son of a Bostonian soap- and candle-maker, Benjamin Franklin was 10 years old when his formal education ended, but his naturally enterprising spirit saw him thrive in multiple occupations, from writer and politician to inventor and scientist. His professional life began at the age of 12, when he was apprenticed to his older brother, who was a printer.

Franklin had an innate aptitude for learning and threw himself into the world of books, reading avidly and improving his writing skills. He became an adept printer and satirical columnist before relocating to Philadelphia, where in 1728, he set up as a printer himself. The financial success he enjoyed was so great that by 1748, he retired from business and applied himself to personal pursuits, including philanthropic projects and politics, but particularly science. He went on to make significant contributions to the field of electricity, for which he became internationally famous.

Experimenting with electricity

In the 1700s, electricity was little understood. It was mainly seen in tricks performed by showmen, and many believed that it was supernatural.

In 1746, Franklin witnessed its effects at a lecture given by the scientist Archibald Spencer and was immediately drawn to discover the science behind it. To that end, he bought Spencer's equipment and conducted his own experiments. His findings led him to conclude that the prevailing theory of electricity—that it was

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While experimenting with various kinds of equipment, Franklin noticed that pointed electrodes were better at conducting electricity through air than spherical ones.

MILESTONES

PRINTING BUSINESS

Sets up as a printer in Philadelphia and becomes the official printer of Pennsylvania in 1730.

BECOMES SCIENTIST

Accrues sufficient wealth by 1748 to retire from business and focus on his scientific experiments.

ELECTRICITY STUDY

Publishes a text on his early experiments in electricity in 1751, and proposes his kite experiment in 1752.

OCEAN MAPPING

Charts and names the Gulf Stream between 1764 and 1765 so ships can avoid it to speed their passage.

“Do not **squander**
Time; for that's the
Stuff **Life is made** of.”

Benjamin Franklin, 1746